EV Market Analytics Dashboard

Tableau Dashboard, Story and Flask Web Integration

Project Documentation

Tools: Tableau Public, SQL (MySQL syntax), Python, Flask, GitHub
Dataset: EV market data, 299,999 vehicle records, 37 columns

1. Problem Understanding

The electric vehicle industry is growing quickly across hatchbacks, sedans, SUVs, buses, trucks and two-wheelers, using battery technologies such as LFP, NMC, NCA, Sodium-Ion and Solid-State. Manufacturers, investors and policymakers need to monitor annual sales, battery capacity, driving range, customer satisfaction, manufacturing distribution and market share. This project builds an interactive Tableau dashboard and story, embedded in a Flask web application, that answers the questions of three stakeholders.

Scenarios and the questions they ask

Stakeholder	Questions
Emily Johnson, EV Market Strategy Director	Which categories contribute most to annual sales? Who are the leading manufacturers? How does customer satisfaction differ by segment? How do battery adoption, driving range and growth compare?
David Chen, Automotive BI Manager	Manufacturer market share, annual sales, battery technology distribution, manufacturing country analysis, customer ratings, and competitive advantages.
Dr. Sarah Williams, Mobility Research Analyst	Comparison of battery technologies, driving range improvements, global manufacturing distribution, and customer satisfaction by category.

2. Data Collection

The source was an archive containing **ev_market_train.csv** and **ev_market_test.csv**. Both files share the same 37 columns and have no overlapping vehicle IDs. They were merged into one file, **ev_market_clean.csv**, with a `dataset_split` column recording the origin, and one duplicate vehicle ID was removed, leaving **299,999 rows**.

Key characteristics

- 200 manufacturers, 15 vehicle categories, 5 battery types and 10 manufacturing countries; model years 2018 to 2026.
- Missing values: annual_sales_units (about 1%), customer_rating (about 2%), battery_cycle_life, resale_value_pct_3yr and infotainment_screen_inch. Tableau ignores blanks in sums and averages, so they were left as they are.
- Fields used most: manufacturer, category, battery_type, annual_sales_units, range_km, battery_capacity_kwh, customer_rating, manufacturing_country, model_year, price_usd.

3. SQL Operations

A MySQL schema (`ev_vehicles` table with indexes on manufacturer, category and model year) and analysis queries were written in `schema_and_sql.sql`. **Note:** the queries were executed on the full 299,999-row dataset using SQLite, which gives the same results as MySQL for these queries; Tableau was then connected directly to the CSV rather than to a live MySQL server. The tables below are the query results.

Query 1 - Sales and average rating by category

Category	Total sales (units)	Avg rating
Two-Wheeler Scooter	259,647,705	4.09
Sedan	231,659,706	4.09
Hatchback	231,033,059	4.08
Compact SUV	229,547,977	4.09
Three-Wheeler	200,227,586	4.09
Crossover	197,079,358	4.09
Luxury Sedan	174,102,587	4.09
Micro City Car	173,292,023	4.09
Full-size SUV	171,655,592	4.09
Luxury SUV	143,466,600	4.08
Van / MPV	143,106,198	4.09
Sports Car	113,930,967	4.09
Electric Truck	19,528,615	4.08
Pickup Truck	19,352,279	4.09
Electric Bus	15,672,798	4.09

Query 2 - Top 10 manufacturers and market share

Manufacturer	Total sales (units)	Share %
AmpereWorks	13,736,241	0.59%
SurgeWorks	13,224,706	0.57%
CircuitWorks	12,870,982	0.55%
WattRides	12,828,785	0.55%
IonAutomotive	12,790,334	0.55%
CurrentMobility	12,756,035	0.55%
ElektraMobility	12,747,265	0.55%
VoltAutomotive	12,705,609	0.55%
VoltMobility	12,704,584	0.55%
VoltWorks	12,697,003	0.55%

Query 3 - Battery technology adoption

Battery	Models	Total sales	Avg range km	5-yr degr. %
NMC	101,508	783,281,459	324.2	14.02
LFP	96,137	744,520,162	324.4	10.12
NCA	51,333	400,863,944	325.0	14.03
Sodium-Ion	29,925	229,088,254	325.5	14.0
Solid-State	21,096	165,549,231	322.9	7.42

Query 4 - Production by manufacturing country

Country	Models	Total sales
China	30,142	237,384,272
India	30,285	235,632,921
Vietnam	30,030	235,339,319
UK	29,884	234,225,926
USA	30,029	232,658,539
South Korea	30,155	232,461,603
France	29,961	230,558,584
Germany	29,916	230,348,229
Sweden	29,837	228,023,679
Japan	29,760	226,669,978

Two further queries (year-over-year growth by category using a window function, and average range by category and battery type) are in the SQL file and the sql_results folder.

4. Tableau Connection and Data Preparation

Tableau was connected with **Connect > Text file** to ev_market_clean.csv and set to use an **Extract** for performance. manufacturing_country was assigned the Country/Region geographic role. Numeric fields were verified as measures and text fields as dimensions.

Calculated fields

Field	Purpose
Total Sales	SUM of annual_sales_units
Market Share %	Share of total sales
Market Share Overall	Share using a FIXED LOD so it stays correct under filters
Avg Range / Avg Rating	AVG of range_km / customer_rating
Rating Band	Excellent / Good / Average / Poor from customer_rating
Price Segment	Budget / Mid / Premium from price_usd
Range per kWh	Range efficiency per unit of battery capacity
Sales YoY %	Year-over-year growth using LOOKUP (table calculation)
Battery Type Share	Percent of total sales by battery type

5. Data Visualizations

Nine visualizations were built, each tied to the scenario questions.

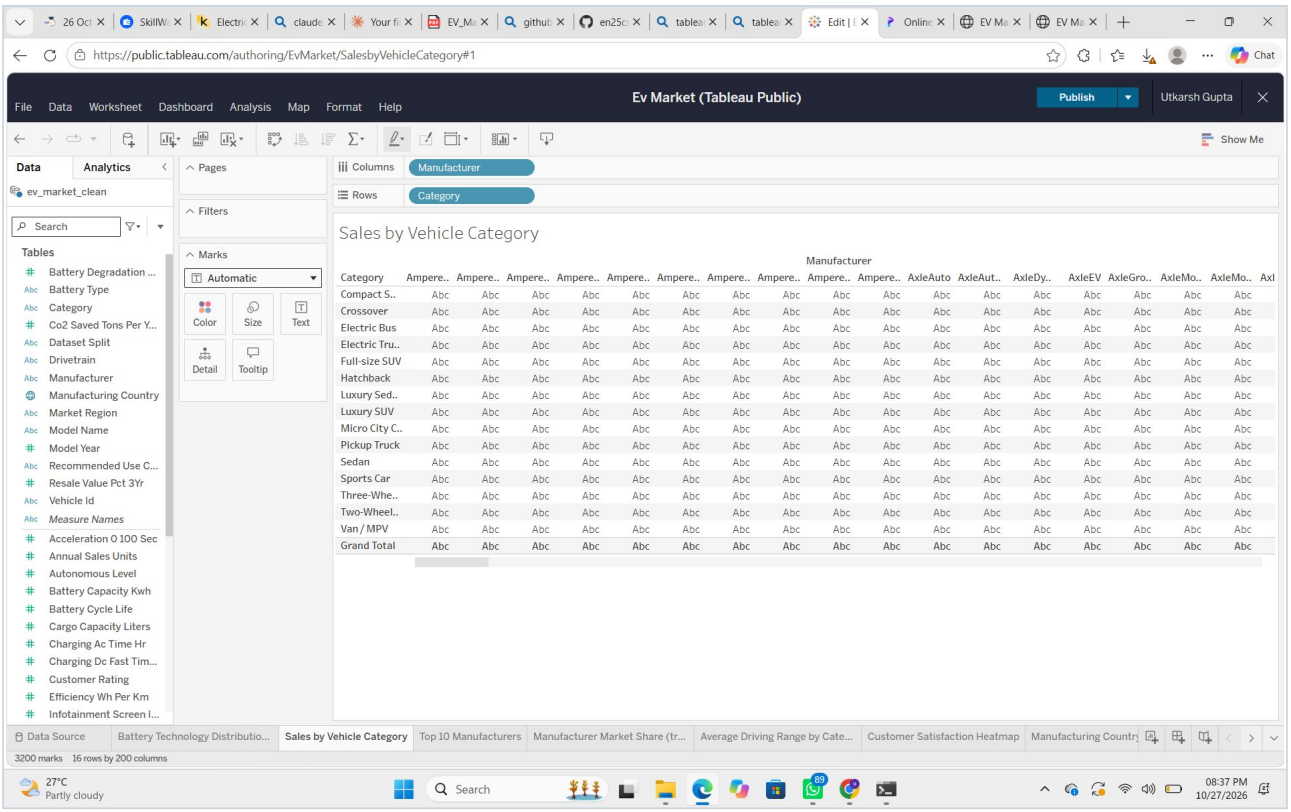
Visualization summary

#	Visualization	Chart type	Answers
V1	Sales by vehicle category	Bar	Emily: categories contributing most to sales
V2	Top 10 manufacturers	Bar (Top 10 filter)	Emily, David: leading manufacturers
V3	Manufacturer market share	Treemap	David: market share
V4	Battery technology distribution	Pie	All: battery adoption
V5	Average driving range by category	Bar + average line	Emily, Sarah: range comparison
V6	Customer satisfaction heatmap	Highlight table	Emily, Sarah: satisfaction by segment
V7	Manufacturing country map	Filled map	David, Sarah: production distribution
V8	Sales trend by category over years	Line + trend line	Emily: high-growth segments
V9	Battery capacity vs driving range	Scatter + trend line	Sarah: range improvements

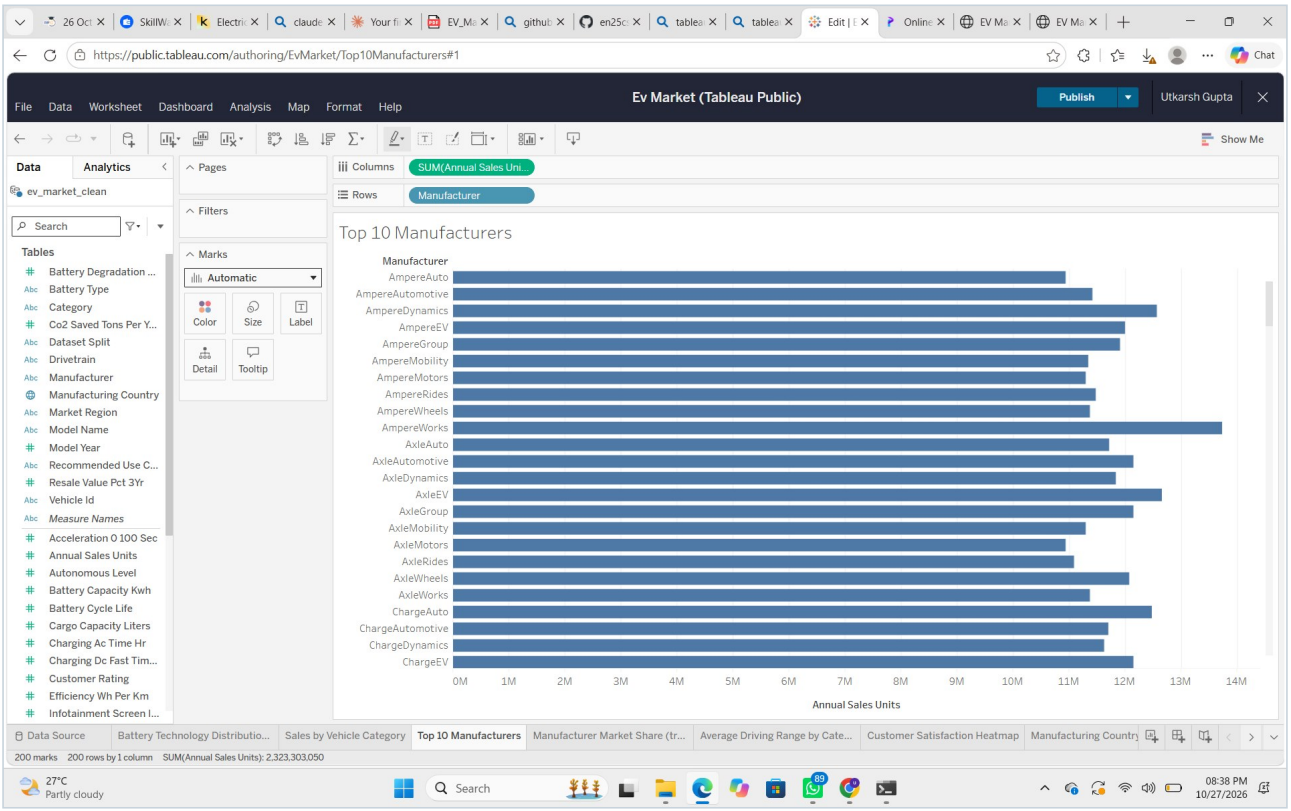
5a. Visualization Screenshots

Screenshots of the nine worksheets built in Tableau Public (authoring view).

V1 - Sales by vehicle category

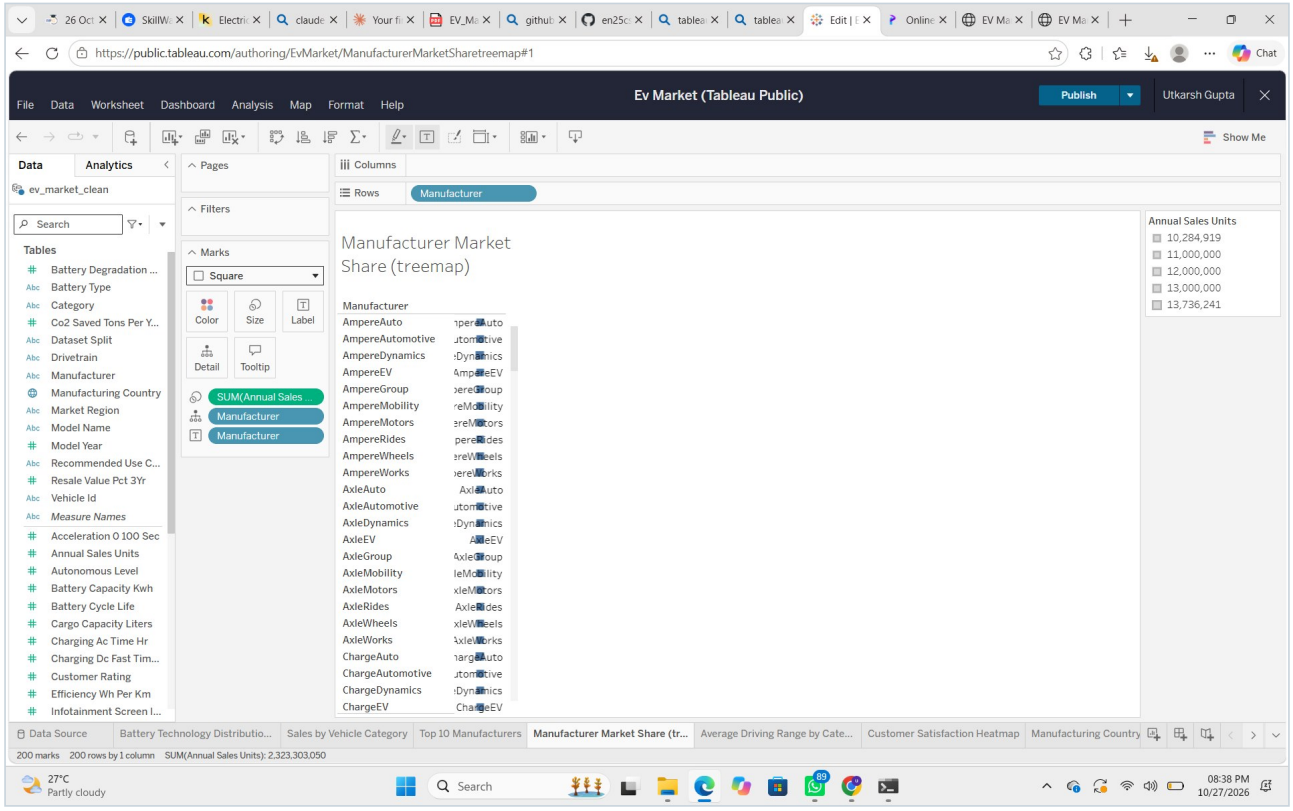


V2 - Top 10 manufacturers

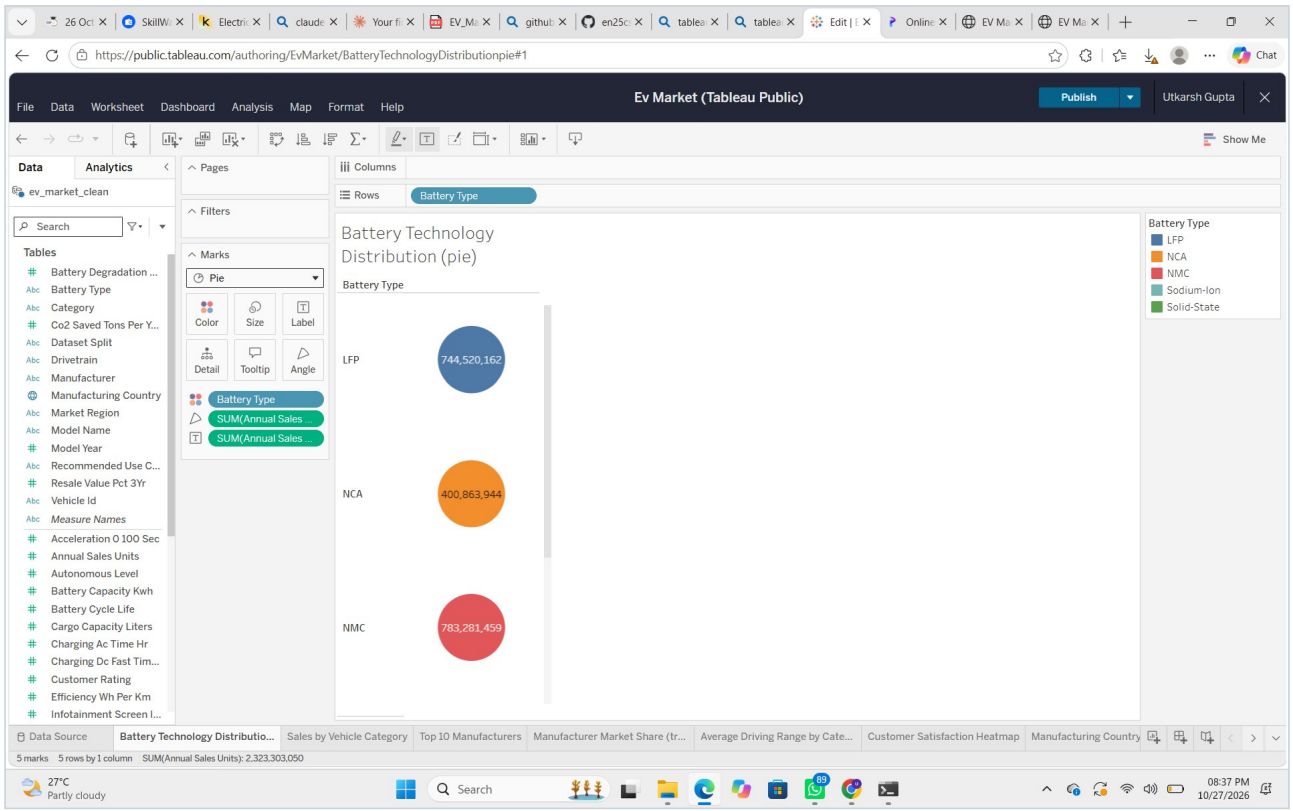


5a. Visualization Screenshots (continued)

V3 - Manufacturer market share (treemap)

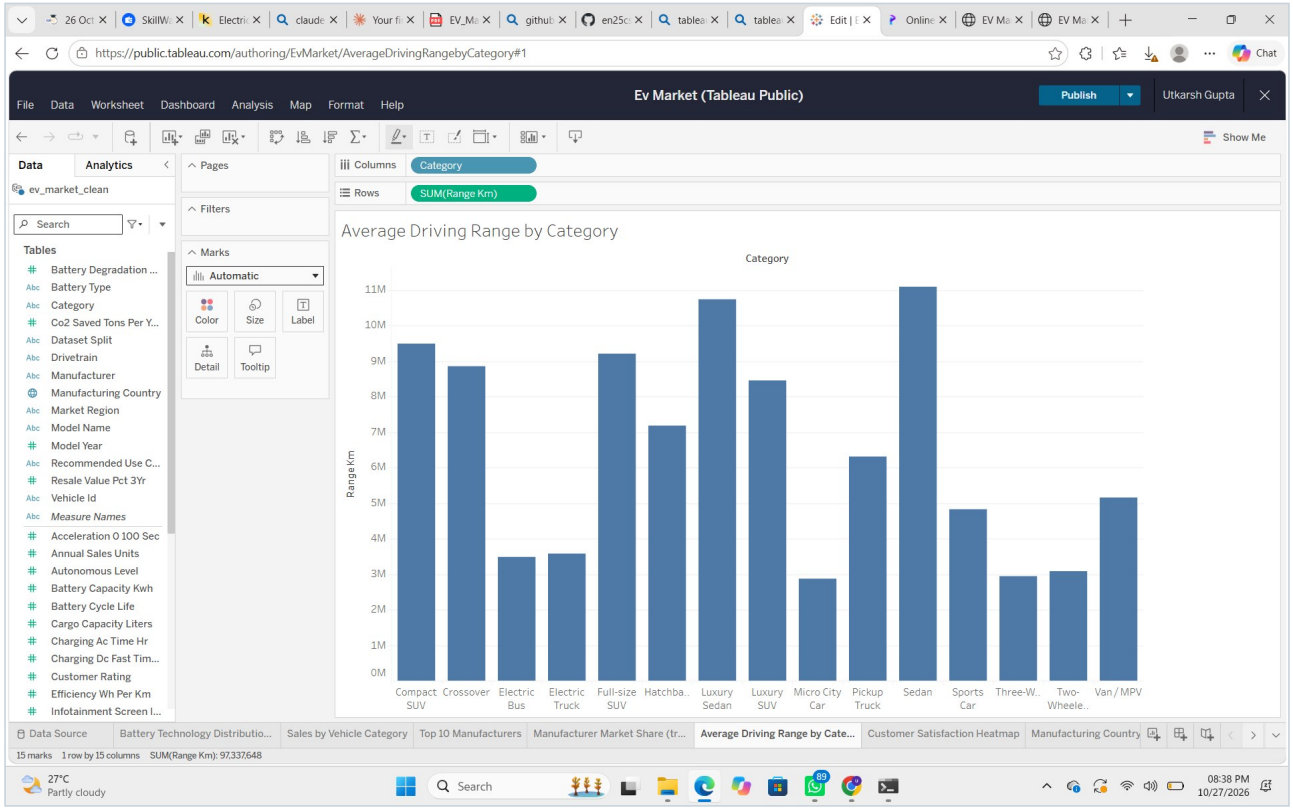


V4 - Battery technology distribution (pie)

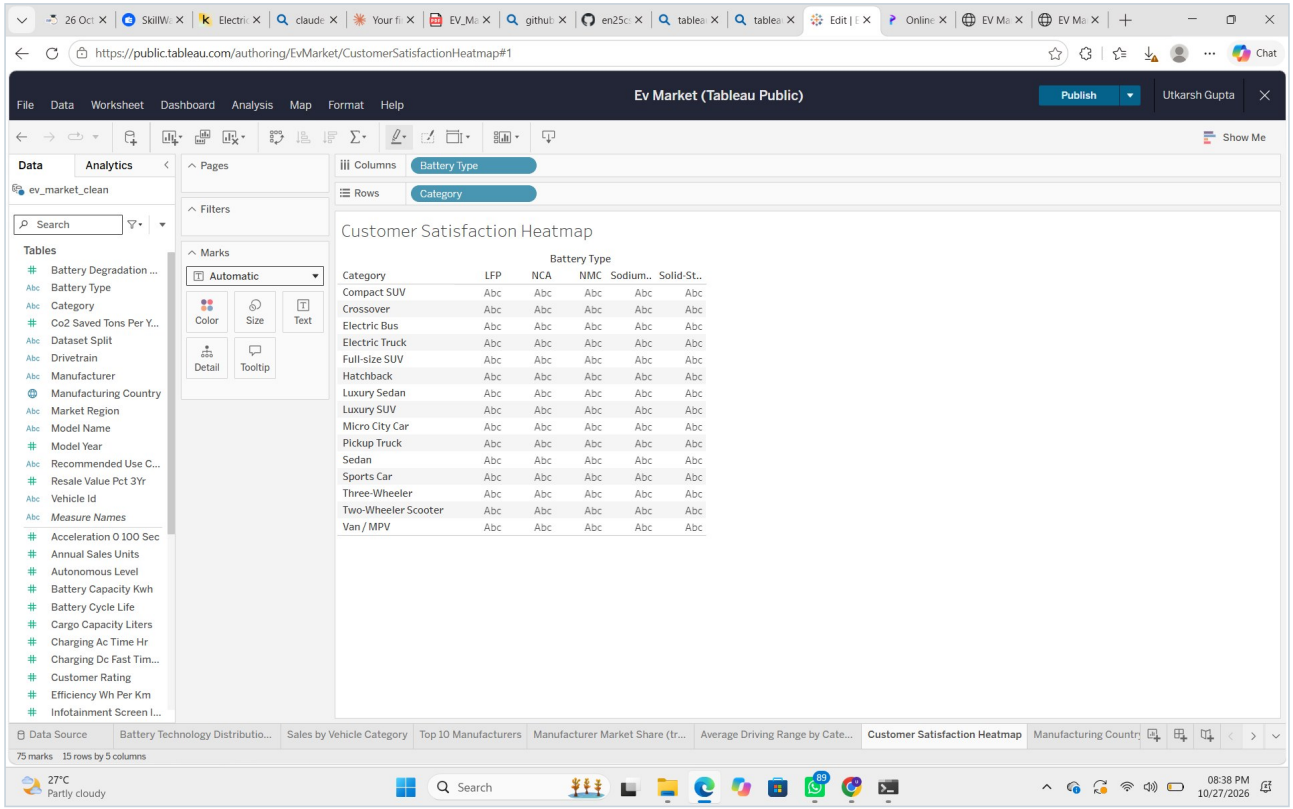


5a. Visualization Screenshots (continued)

V5 - Average driving range by category

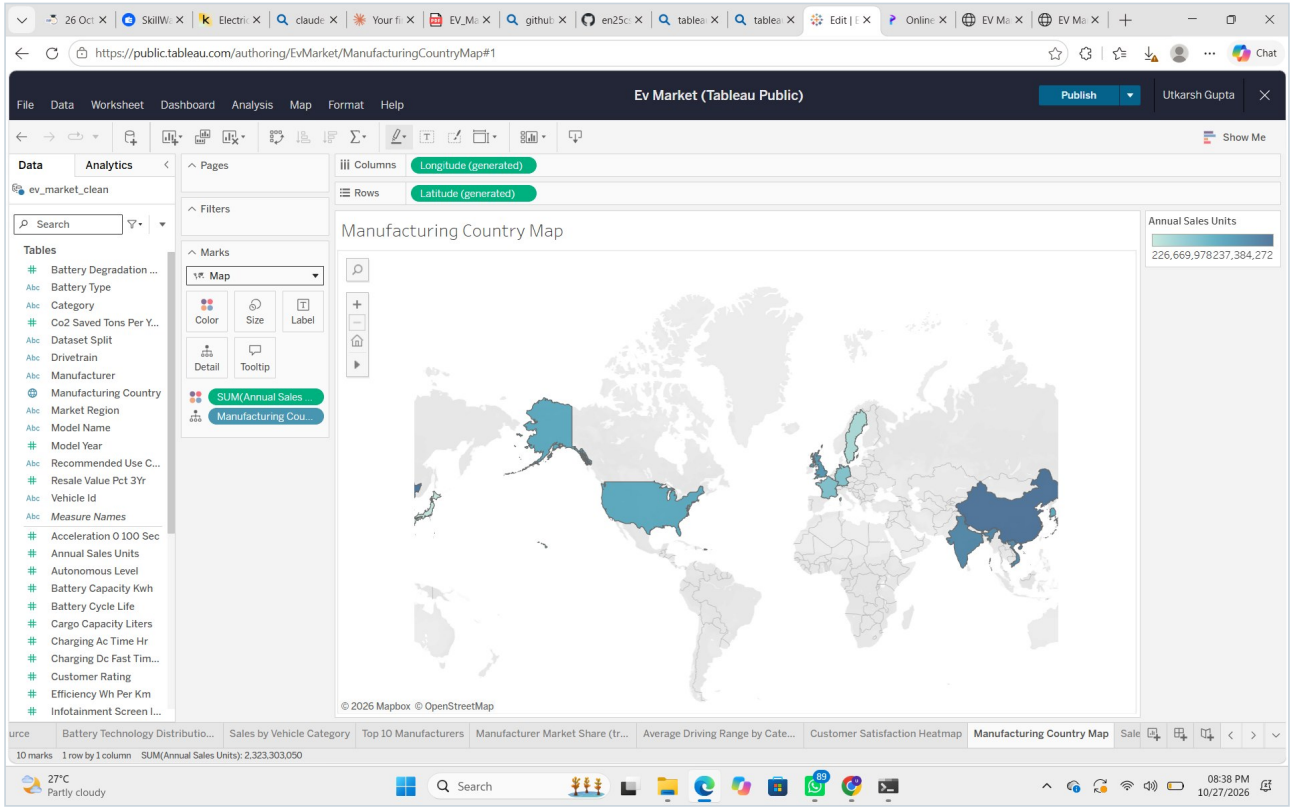


V6 - Customer satisfaction heatmap

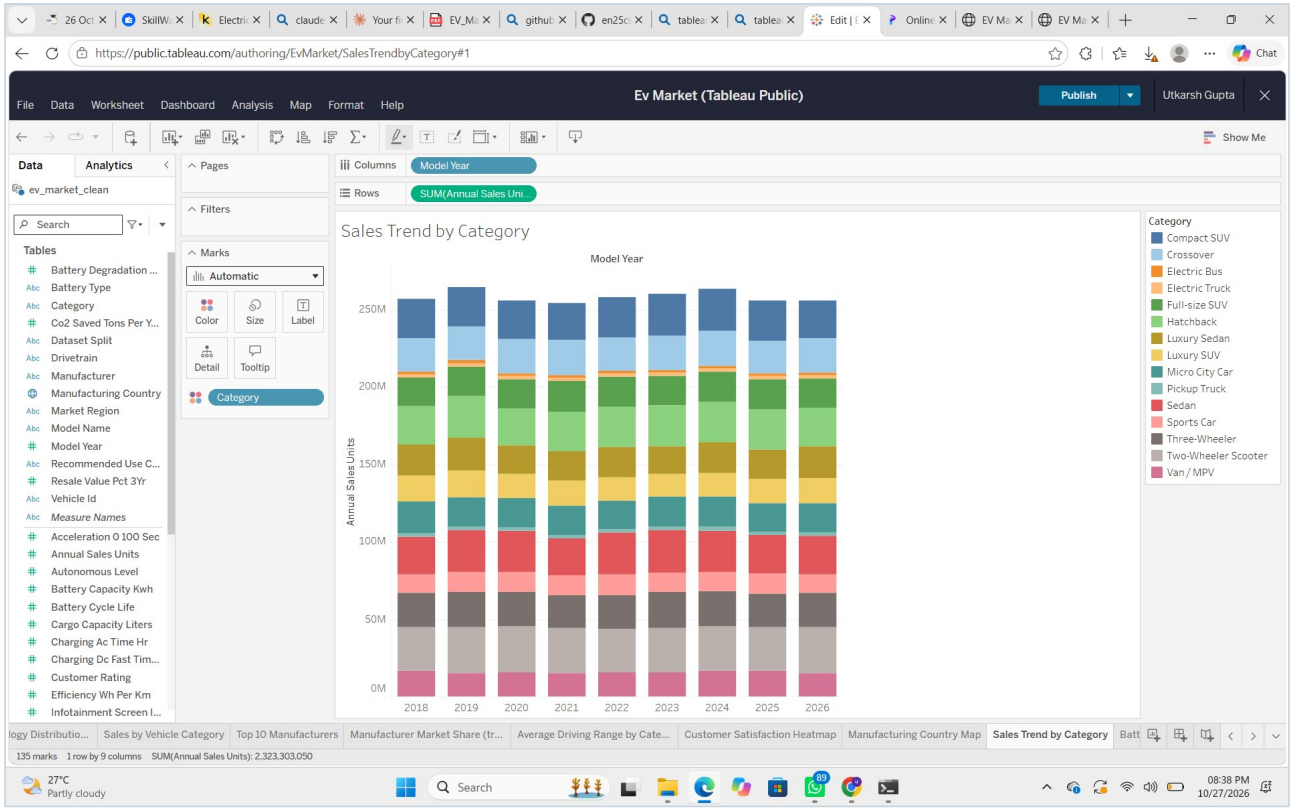


5a. Visualization Screenshots (continued)

V7 - Manufacturing country map

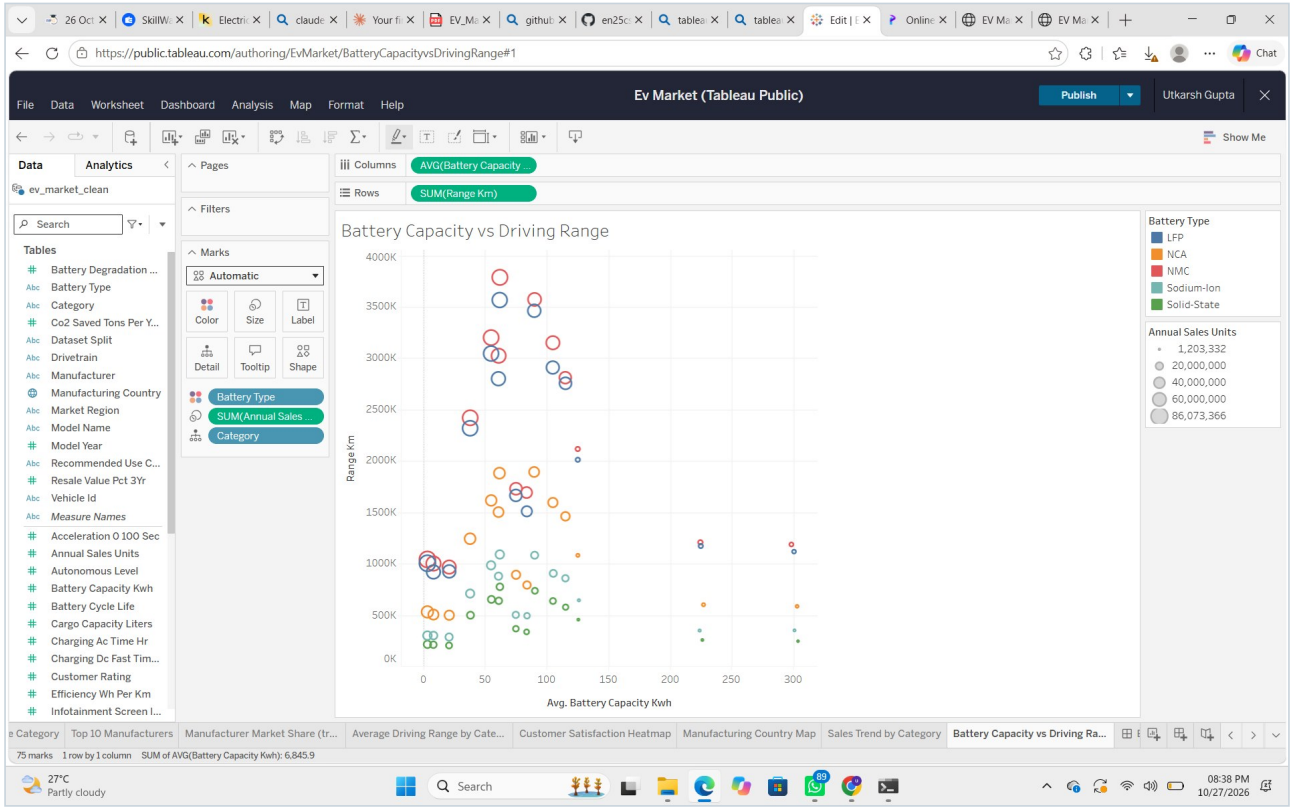


V8 - Sales trend by category



5a. Visualization Screenshots (continued)

V9 - Battery capacity vs driving range



6. Dashboard

The dashboard uses an Automatic size layout with the nine charts in three rows. Interactive filters apply to all worksheets: model year, category, market region, battery type, manufacturing country, price segment and rating band. Charts V1, V3, V4 and V7 act as filters for the others, and a phone layout was added. The manufacturer filter is added to context so the Top 10 charts respond correctly.

7. Story

The story has four scenes: Market Overview (V1, V2, V8), Competitive Landscape (V3, V7, V4), Technology and Satisfaction (V5, V6, V9) and Recommendations (full dashboard).

8. Performance Testing

- Amount of data: 299,999 rows using a Tableau Extract.
- Filters: 7 interactive filters.
- Calculated fields: 9.
- Visualizations: 9 worksheets, 3 dashboards, 1 story with 4 scenes.

Performance Recording (Help > Settings and Performance) was run while applying filters.

9. Web Integration with Flask

A Flask app (`app.py`) serves two routes, `/` for the dashboard and `/story` for the story. Both pages extend a shared `base.html` template with a navigation bar and embed the published Tableau Public views through the Tableau JavaScript API.

To run: open a terminal in the flask_app folder, run `pip install -r requirements.txt` then `python app.py`, and open http://127.0.0.1:5000.

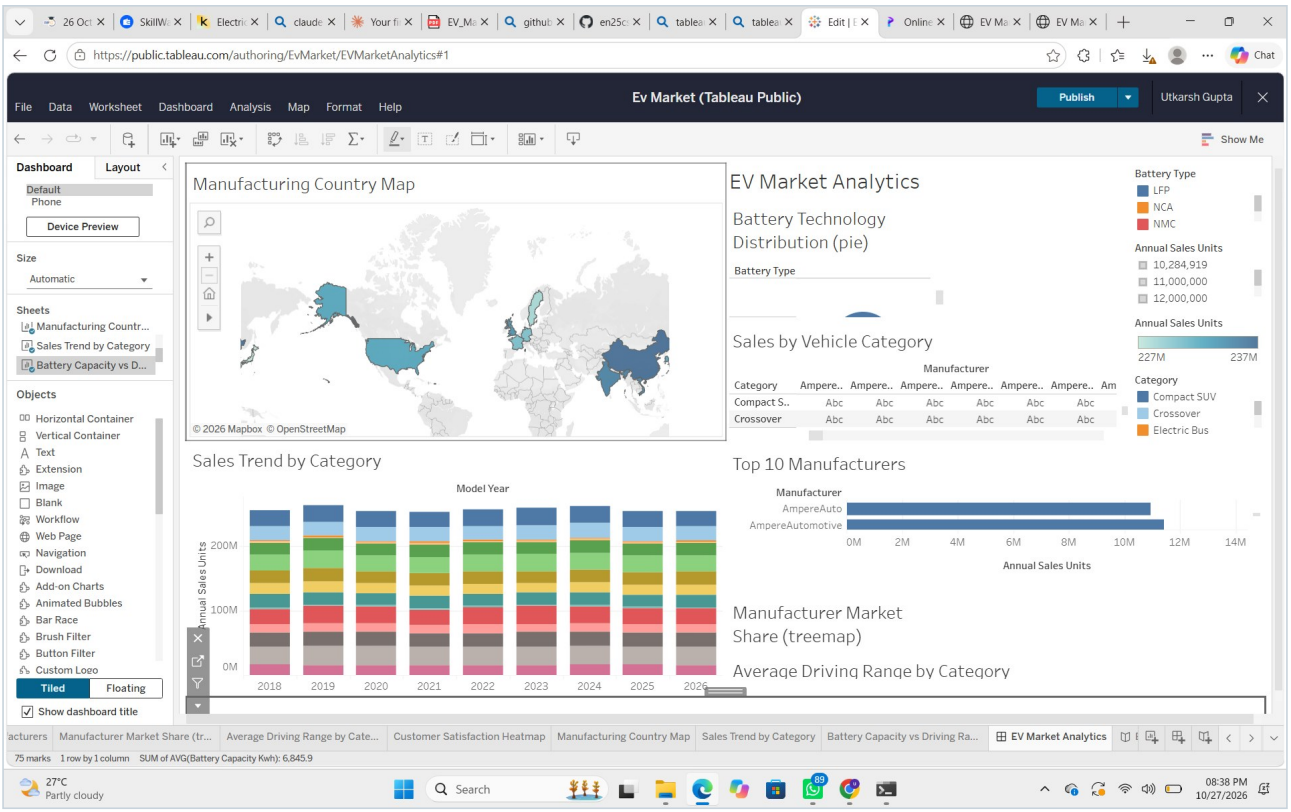
10. Key Findings

- **Categories:** Two-Wheeler Scooters lead with about 259.6M units, followed by Sedans (231.7M), Hatchbacks (231.0M) and Compact SUVs (229.5M). Electric Buses, Pickup Trucks and Electric Trucks are lowest, at about 15.7M to 19.5M.
- **Manufacturers:** the market is highly fragmented. The leader, AmpereWorks, holds only about 0.59% share, and the top 10 are all within 0.55% to 0.59%.
- **Battery technology:** NMC (783M units) and LFP (745M) lead adoption. Solid-State has the lowest 5-year degradation (7.4%, against about 14% for NMC, NCA and Sodium-Ion) but the smallest sales volume.
- **Geography:** production is spread almost evenly across 10 countries (227M to 237M units each), with China slightly ahead.
- **Range and satisfaction:** average range is similar across battery types (about 323 to 325 km) and average customer rating is about 4.08 to 4.09 everywhere, so these measures do not separate segments strongly in this dataset.

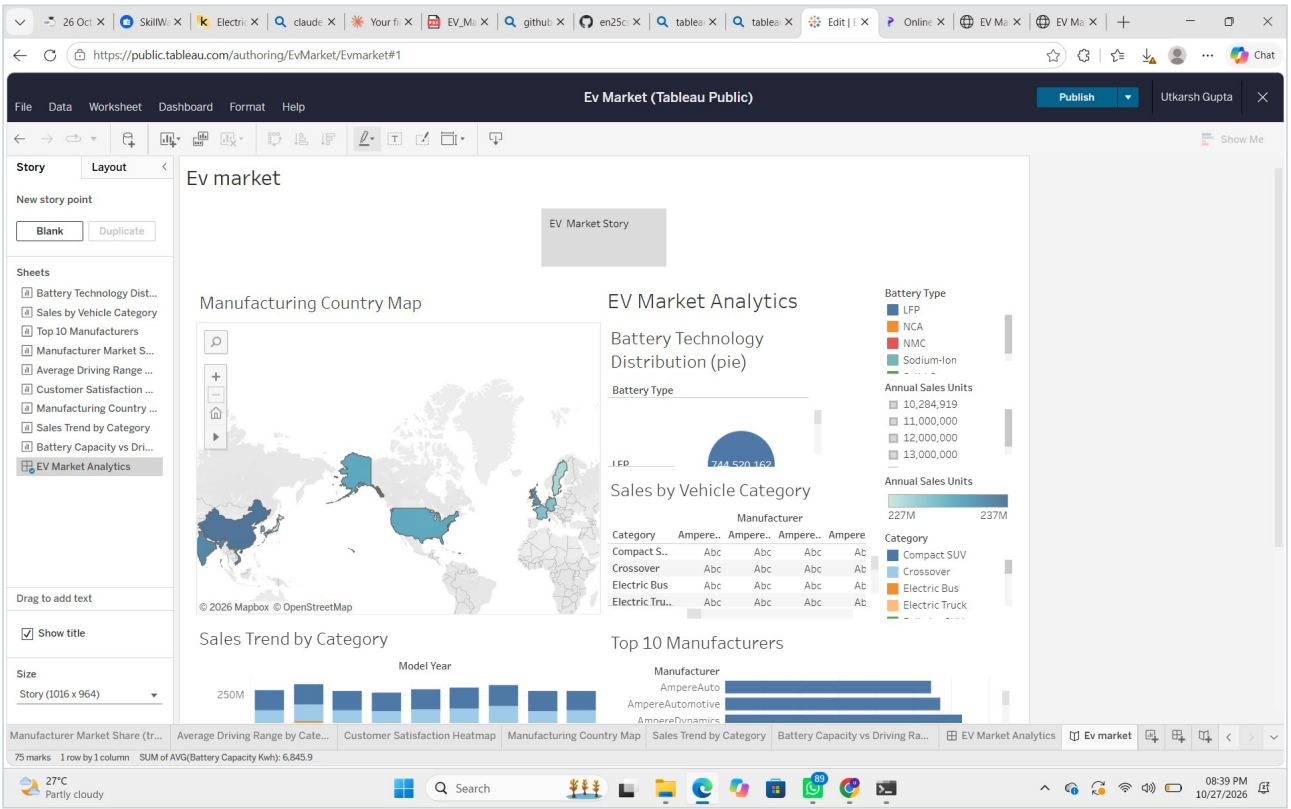
6a. Dashboard and Story Screenshots

Dashboard and story as built in Tableau Public (authoring view).

Dashboard - EV Market Analytics



Story - EV Market



11. Conclusion and Recommendations

The dashboard gives each stakeholder a direct answer: Emily can see that two-wheelers and mainstream cars drive sales; David can see that no single manufacturer dominates, so share is won through segment and region positioning; and Sarah can see that Solid-State batteries offer a durability advantage while range and ratings are similar across technologies. Entrants should consider the high-volume two-wheeler and sedan segments, and invest in battery durability as a differentiator.

12. Project Links

- Tableau Public dashboard:
  https://public.tableau.com/shared/YY223967S?:display_count=n&:origin=viz_share_link
- Tableau Public story:
  https://public.tableau.com/shared/TWJ8T86FW?:display_count=n&:origin=viz_share_link
- GitHub repository:
  https://github.com/en25cs3011010-eng/EV-market-Dashboard.git
- Demo video: *add your video link*

Appendix A. Website Source Code (ev_website_preview.html)

This page embeds the published Tableau Public dashboard (shared/YY223967S) and story (shared/TWJ8T86FW) with Dashboard and Story tabs. It is the standalone version of the Flask templates dashboard.html and story.html.

```
<!doctype html><html lang="en"><head><meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>EV Market Analytics</title>
<style>
body{margin:0;font-family:Segoe UI,Arial,sans-serif;background:#f4f6f8;color:#1b2a3a}
nav{background:#0b3d5c;padding:14px 24px;display:flex;gap:24px;align-items:center}
nav h1{color:#fff;font-size:20px;margin:0;flex:1}
nav a{color:#bfe3f5;text-decoration:none;font-weight:600;cursor:pointer}nav a:hover,nav
    a.active{color:#fff;border-bottom:2px solid #fff}
.wrap{max-width:1300px;margin:20px auto;padding:0 16px}
.card{background:#fff;border-radius:10px;box-shadow:0 2px 8px
    rgba(0,0,0,.08);padding:12px;overflow-x:auto}
.hidden{display:none}
</style></head><body>
<nav><h1>EV Market Analytics</h1>
<a id="t1" class="active" onclick="show(1)">Dashboard</a><a id="t2" onclick="show(2)">Story</a></nav>
<div class="wrap"><div class="card">
<div id="p1"><div class='tableauPlaceholder' id='vizDashboard' style='position:relative;width:100%'>
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    value='3' />
<param name='path' value='shared/YY223967S' /><param name='toolbar' value='yes' /><param
    name='display_count' value='n' /><param name='origin' value='viz_share_link' /></object></div></div>
<div id="p2" class="hidden"><div class='tableauPlaceholder' id='vizStory'
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    name='display_count' value='n' /><param name='origin' value='viz_share_link' /></object></div></div>
</div></div>
<script>
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document.getElementById('t1').classList.toggle('active',n===1);document.getElementById('t2').classList.to
    ggle('active',n===2);}
['vizDashboard','vizStory'].forEach(function(id){var
    d=document.getElementById(id),o=d.getElementsByTagName('object')[0];
var s=document.createElement('script');s.src='https://public.tableau.com/javascripts/api/viz_v1.js';o.par
    entNode.insertBefore(s,o);});
</script></body></html>
```